EGIRE ROBOTICS

Explore Engineer Excel

FOR ENGINEERING STUDENTS AND GRADUATES

BUILD IT. FLY IT. FIX IT.

A hands-on FPV drone engineering programme. In 50 days you progress from your first solder joint to a drone that you built, tested and flew yourself.



50	5	2	100
DAYS OF TRAINING	HANDS-ON LABS	FPV DRONES BUILT: ANALOG AND DIGITAL	PAGE LAB TEXTBOOK

HANDS-ON ENGINEERING

LEARN BY BUILDING AND FLYING

SOLDERING

Through-hole and surface-mount soldering, rework, motor and ESC wiring. Every joint is inspected and graded.

BUILD

Assemble the frame, flight controller, motors, camera, video transmitter and radio. Build one analog and one digital 5-inch FPV drone.

FLY

Start in the simulator, then fly line-of-sight and with goggles on analog and digital, with GPS return and missions.

DEBUG

Trace faults step by step using an oscilloscope, a logic analyser, serial output and flight logs.

TROUBLESHOOT

Use a clear method and test tools to find and fix faults in motors, video, radio links and arming.

CODE AND TUNE

Program microcontrollers in C, log data with Python, and tune PID and filters using real flight data.





50 DAYS. 5 PHASES. 5 LABS.

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1–10 DAYS	**ELECTRONICS FOUNDATIONS** Ohm's law, MOSFETs, soldering, oscilloscope and multimeter, power electronics, PCB basics.	LAB 1 Power board
11–20 DAYS	**EMBEDDED SYSTEMS AND PROGRAMMING** C programming, timers and PWM, ADC, UART, SPI, I^2C, IMU sensor fusion, Python logging, Git.	LAB 2 IMU logger
21–30 DAYS	**DRONE ENGINEERING** Flight physics, brushless motors, ESCs, batteries, flight controllers, ELRS radio, PID, filters.	LAB 3 Thrust stand
31–40 DAYS	**BUILD ANALOG AND DIGITAL FPV DRONES** Plan, assemble, solder, wire, configure Betaflight, bench-test, first hover and tuning.	LAB 4 Build and tune
41–50 DAYS	**PILOTING AND ADVANCED SYSTEMS** Simulator, FPV flying, regulations, GPS rescue, missions, computer vision, final assessment.	LAB 5 Race + mission

WHY STUDENTS CHOOSE IT

- **Hands-on first.** A lab every day: measure, solder, build and fly.
- **Real engineering.** Calculations, test data and engineering reports.
- **Fault-finding skills.** Use an oscilloscope, logic analyser and flight logs to find and fix faults.
- **Safety first.** Trainer-supervised flying, checklists and clear rules.
- **A complete course textbook.** 100 pages written for this programme: a lesson for every day, worked examples, labs, formula sheets and an answer key.
- **Mentor support.** Guidance on projects, your portfolio and further study.

WHY DRONE ENGINEERING

ONE MACHINE. EVERY SKILL.

A drone brings electronics, programming, mechanics, aerodynamics, radio and control together in a single aircraft. Building and flying one is the most direct way to apply engineering.

Notes-based learning. Every day follows the same pages in your textbook: objectives, key equations, a worked example, a lab and check questions. You always know what to learn and how to practise it.

DAILY NOTES

WORKED EXAMPLES | LABS

WHAT YOU TAKE HOME

CERTIFICATE
EgireRobotics certificate on completing the course

LAB LOGBOOK
50 days of measurements, photos and reflections

PORTFOLIO
Git repository, project notes and flight video

SKILLS
solder, build, fly, debug, troubleshoot, code

JOIN THE NEXT BATCH

~~₹35,000~~ **₹24,999** COMPLETE 50-DAY TRAINING FEE

CONTACT US

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Open to engineering students and graduates. A basic grounding in school-level maths and physics is helpful; no prior drone experience is required. Every flight is supervised and follows local drone rules. Dates and seat availability are confirmed by your trainer.